

Laboratory Work No. 2

Creating and Populating a Relational Database

Objective:

Learn how to create tables and insert data into a relational database using the system you installed in Lab 1.

Introduction:

After installing your Relational Database Management System (RDBMS) in the previous lab, it's time to build a complete database. In this lab, you will create a university organization database based on the schema provided below. You will also practice different methods of inserting data into the database and evaluate their efficiency.

Database Schema:

department(dept-name, building, budget)
instructor(ID, name, dept-name, salary)
student(ID, name, dept-name, tot-cred)
advisor(s-ID, i-ID)
course(course-id, title, dept-name, credits)
teaches(ID, course-id, sec-id, semester, year)
takes(ID, course-id, sec-id, semester, year, grade)
classroom(building, room-number, capacity)
time-slot(time-slot-id, day, start-time, end-time)
section(course-id, sec-id, semester, year, building, room-number, time-slot-id)
prereq(course-id, prereq-id)

Task:

1. Create a new database in your chosen RDBMS (from Lab 1).
2. Define all tables according to the schema above, using appropriate data types and constraints.
3. Insert sample data into each table using at least one of the following methods:
 - i. Manual SQL statements (e.g., INSERT INTO ...)
 - ii. Bulk import tools or CSV import features
4. Test your database by running a few simple queries (e.g., list all students or instructors).
5. Compare the different data insertion methods and explain which one you find most efficient and why.

Deliverables:

1. A short **report** describing your process, including challenges and how you solved them.
2. **Screenshots** showing key steps (table creation, data insertion, sample queries).
3. **Upload** the database file or SQL script you created (e.g., .db for SQLite or .sql for MySQL).

Good luck!